## A Contributions

The contributions of this study are as follows:

- **Introduction of the SOLAR 10.7 Billion-Parameter Model**: We have released the SOLAR 10.7B model, which is not only depth-wise scaled but also continually pretrained. The availability of SOLAR 10.7B under the Apache 2.0 license permits commercial usage, enabling the integration of this advanced model into a diverse range of products and services. This bridges the gap between academic research and practical applications, fostering wider accessibility and utility in various fields.
- **Superior Performance Across Diverse Benchmarks**: SOLAR 10.7B excels in various benchmarks, outperforming established models like Llama 2 and Mistral 7B in reasoning, mathematics, and the MMLU framework.
- **Advancement in Instruction-Following Capabilities**: The introduction of SOLAR 10.7B-Instruct, a variant fine-tuned for enhanced instruction-following abilities, marks a significant improvement in the model's ability to understand and execute complex instructions.

Dahyun Kim, Chanjun Park, Sanghoon Kim, and Wonsung Lee contributed equally to this paper. Sanghoon Kim led the Foundation Model part, with Dahyun Kim, Wonho Song, Yunsu Kim, and Hyeonwoo Kim. Chanjun Park led the Data and Evaluation (Data-Centric LLM) part, with Yungi Kim, Jihoo Kim, Changbae Ahn, Seonghoon Yang, Sukyung Lee, and Hyunbyung Park. Wonsung Lee led the Adaptation Modeling part, with Gyoungjin Gim, Hyeonju Lee, and Mikyoung Cha. Hwalsuk Lee performed the role of the overall project operation. All these individuals contributed to the creation of SOLAR 10.7B.

## B Related Works and Background

### B.1 Large Language Models

Following the advent of context-based language models, various studies have revealed a "scaling law" (Kaplan et al., 2020; Hernandez et al., 2021; Anil et al., 2023), demonstrating a positive correlation between the size of model and training data and model performance. This has led to the emergence of Large Language Models (LLMs). Unlike previous language models, LLMs possess the ability for In-context learning, including Zero-shot learning (Radford et al., 2019) and Few-shot learning (Brown et al., 2020), allowing them to perform new tasks without updating model weights. These capabilities of LLMs, not evident in smaller models, are referred to as Emergent abilities (Wei et al., 2022a).

### B.2 Mixture of Experts

In the landscape of machine learning architectures, the Mixture of Experts (MoE) models like (Shazeer et al., 2017; Shen et al., 2019; Komatsuzaki et al., 2022) has gained attention for its capability to address the challenges posed by complex and heterogeneous data. MoE models offer notable benefits, including enhanced output diversity, allowing for the capture of intricate patterns within the input space. Moreover, their computational efficiency, especially when implemented in a sparse form, has made them valuable in scenarios where resource constraints are a consideration (Shazeer et al., 2017; Komatsuzaki et al., 2022).

However, efficient implementation of MoE models poses a considerable challenge, primarily due to the intricacies associated with dynamic routing and load-imbalanced computation (Gale et al., 2023). Existing hardware and software for deep learning, such as TPUs and XLA compilers, often demand static knowledge of tensor shapes, making MoE implementation on TPU challenging.

While GPU implementation offers more flexibility, sparse computation compatibility becomes a hurdle. Striking the right balance between fixing the size of each expert to facilitate efficient computation and maintaining model quality creates a tradeoff between information preservation and hardware efficiency. This tradeoff, in turn, necessitates careful consideration during hyperparameter tuning, adding a layer of complexity to the implementation of MoE models, potentially offsetting their advantages. Given the formidable challenges in MoE model implementation, it becomes almost inevitable for researchers and practitioners to resort to specialized tools and frameworks, such as Tutel (Hwang et al., 2023) or Megablocks (Gale et al., 2023).

Departing from the horizontal expansion characteristic of MoE models, the DUS method introduces model scaling in the vertical dimension. Notably, DUS does not introduce dynamism in the scaled model, which significantly reduces the com-